QF605 Fixed Income Securities

G2 Group Project Report

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Part I - Bootstrapping Swap Curves

1.1 OIS Discount Factor

Overnight Index Swaps (OIS) pay a fixed swap rate vs. a floating leg whose rate is compounded according to the daily overnight rate. OIS are quoted by their par swap rate.

Using the given OIS data, which consists of tenors ranging from 6m to 30y with annual payment, we have computed the OIS discount factor $D_o(0,T)$ and plotted the discount curve for $T \in [0,30]$ in Table 1 and Figure 1.1 respectively.

At the time of the swap's inception $PV_{fix} = PV_{flt}$, and we have utilized this condition to subsequently bootstrap the OIS discount factors.

$$\sum_{i=1}^{n} D_o(0,T_i) \times \Delta_{i-1} \times OIS = \sum_{i=1}^{n} D_o(0,T_i) \times \Delta_{i-1} \times L(T_{i-1},T_i)$$

For maturities $\leq$ a year, the discount factors are calculated as follows:

- At 6m, $\Delta_{6m}= 0.5$ and $L(0,6m) = \left(1+\frac{f_0}{360}\right)^{180} - 1$, f_0 is the daily compounded overnight rate for $[0,6m]$
- At 1y, $\Delta_{1y}= 1$ and $L(0,1y) = \left(1+\frac{f_0}{360}\right)^{180}\left(1+\frac{f_1}{360}\right)^{180} - 1$, f_1 is the daily compounded overnight rate for the period of $[6m, 1y]$

For longer maturities, $\Delta_{i-1}= 1$, where f_i is the daily compounded overnight rate over$[T_{i-1},T_i]$,

$$L(T_{i-1},T_i) = \left(1+\frac{f_i}{360}\right)^{360} - 1$$

We subsequently solved for f_i, and bootstrapped the relevant OIS discount factors. For the year gaps between tenors, we carried out linear interpolation and assumed overnight rates to be piecewise constant. See Table 1.

Figure 1.1 : OIS Discount Curve

1.2 LIBOR Discount Factor

Unlike OIS rates, LIBOR rates are uncollateralized and hence, discount factors will be higher than those computed using OIS rates. Given LIBOR and IRS data with tenors ranging from 6m to 30y with semi-annual payment frequency, we computed the relevant LIBOR discount factors $D(0,T)$ and plotted the discount curve for $T \in [0,30]$ in Table 1 and Figure 1.2 respectively

We adopted steps similar to the above for $PV_{fix} = PV_{flt}$, to bootstrap the relevant LIBOR discount factors. However, the discount factors adopted in PV_{fix} and PV_{flt} computation are OIS discount rates, i.e. $D_o(0,T_i)$, where semi-annual OIS discount factors are computed based on backfilled f_i.

The formula for bootstrap LIBOR discount rates is as per below:

$$\sum_{i=1}^{n} D_o(0,T_i) \times \Delta_{i-1} \times IRS = \sum_{i=1}^{n} D_o(0,T_i) \times \Delta_{i-1} \times L(T_{i-1},T_i),$$

$$where\ L(T_{i-1},T_i) = \left[\frac{D(0,T_{i-1}) - D(0,T_i)}{D(0,T_i)}\right]$$

Table 1: Combined Discount Factors (OIS & Discount Factor)

Years	Rate	OIS_df	LIBOR_df
0.5	0.0250	0.9988	0.9877
1.0	0.0280	0.9970	0.9726
1.5	0.0290	0.9953	0.9574
2.0	0.0300	0.9935	0.9422
2.5	0.0308	0.9918	0.9263
3.0	0.0315	0.9900	0.9105
3.5	0.0320	0.9881	0.8947
4.0	0.0325	0.9861	0.8790
4.5	0.0328	0.9842	0.8640
5.0	0.0330	0.9822	0.8490

Years	Rate	OIS_df	LIBOR_df
5.5	0.0335	0.9797	0.8328
6.0	0.0340	0.9773	0.8166
6.5	0.0345	0.9749	0.8004
7.0	0.0350	0.9724	0.7842
7.5	0.0353	0.9697	0.7690
8.0	0.0357	0.9669	0.7537
8.5	0.0360	0.9642	0.7385
9.0	0.0363	0.9615	0.7232
9.5	0.0367	0.9587	0.7080
10.0	0.0370	0.9560	0.6927

Years	Rate	OIS_df	LIBOR_df
10.5	0.0373	0.9531	0.6785
11.0	0.0376	0.9503	0.6644
11.5	0.0379	0.9475	0.6502
12.0	0.0382	0.9446	0.6361
12.5	0.0385	0.9418	0.6219
13.0	0.0388	0.9390	0.6077
13.5	0.0391	0.9361	0.5936
14.0	0.0394	0.9333	0.5794
14.5	0.0397	0.9304	0.5652
15.0	0.0400	0.9276	0.5511

Years	Rate	OIS_df	LIBOR_df
15.5	0.0405	0.9249	0.5368
16.0	0.0410	0.9221	0.5225
16.5	0.0415	0.9194	0.5082
17.0	0.0420	0.9166	0.4939
17.5	0.0425	0.9138	0.4796
18.0	0.0430	0.9111	0.4654
18.5	0.0435	0.9083	0.4511
19.0	0.0440	0.9056	0.4368
19.5	0.0445	0.9028	0.4225
20.0	0.0450	0.9001	0.4082

Years	Rate	OIS_df	LIBOR_df
20.5	0.0453	0.8974	0.3990
21.0	0.0455	0.8948	0.3898
21.5	0.0458	0.8922	0.3805
22.0	0.0460	0.8895	0.3713
22.5	0.0463	0.8869	0.3621
23.0	0.0465	0.8843	0.3529
23.5	0.0468	0.8816	0.3436
24.0	0.0470	0.8790	0.3344
24.5	0.0473	0.8764	0.3252
25.0	0.0475	0.8737	0.3159

Years	Rate	OIS_df	LIBOR_df
25.5	0.0478	0.8711	0.3067
26.0	0.0480	0.8685	0.2975
26.5	0.0483	0.8658	0.2883
27.0	0.0485	0.8632	0.2790
27.5	0.0488	0.8606	0.2698
28.0	0.0490	0.8579	0.2606
28.5	0.0493	0.8553	0.2513
29.0	0.0495	0.8527	0.2421
29.5	0.0498	0.8500	0.2329
30.0	0.0500	0.8474	0.2237

Insights

The table of discount factors clearly highlights the divergence between the OIS and LIBOR discount curves across different maturities. The OIS discount factors are consistently higher than their LIBOR counterparts, reflecting the lower credit and liquidity risk embedded in the OIS curve, which is collateralized. This spread is particularly pronounced at the longer end of the curve illustrating the cumulative effect of even a small spread in continuously compounded rates over time. These differences underscore the importance of using the correct curve depending on the collateralization structure of the derivative being valued.

Figure 1.2: LIBOR Discount Curve

1.3 Forward Swap Rate

We computed 2 scenarios of forward swap rates – first using LIBOR discount rates (uncollateralized) and second using OIS discount rates (collateralized) based on the formulas below. The results are shown below in Table 2.

LIBOR forward swaps

$$\sum_{i=s+1}^{N} D_o(0,T_i) \times \Delta_i \times IRS_{forward} = \sum_{i=s+1}^{N} D_o(0,T_i) \times \left[\frac{D(0,T_{i-1}) - D(0,T_i)}{D(0,T_i)}\right]$$

OIS forward swaps

$$OIS_{forward} = \frac{D_o(0,T_s) - D_o(0,T_N)}{\Delta_i \sum_{i=s+1}^{N} D_o(0,T_i)}$$

Table 2: Forward Swap Rates

Start at	Tenors: 1Y	2Y	3Y	5Y	10Y
1Y	0.0320	0.0333	0.0340	0.0353	0.0384
5Y	0.0393	0.0401	0.0401	0.0411	0.0436
10Y	0.0422	0.0431	0.0441	0.0462	0.0535

Part II – Swaption Calibration

2.1 Displaced Diffusion (DD) Model

Sigma parameters were obtained from the ATM vols of the various swaptions. These values were then used to obtain the beta parameters which would minimise the error given between the lognormal IVs and the black IVs. The key parameters obtained are as follows:

Table 3: Black Implied Volatilities

Start at	Tenors: 1Y	2Y	3Y	5Y	10Y
1Y	0.2250	0.2872	0.2978	0.2607	0.2447
5Y	0.2726	0.2983	0.2998	0.2660	0.2451
10Y	0.2854	0.2928	0.2940	0.2674	0.2437

Table 4: Beta(β) Parameters

		Tenors				
		1Y	2Y	3Y	5Y	10Y
Start at	1Y	1.638e-10	1.017e-7	8.475e-13	8.331e-11	2.500e-5
	5Y	8.974e-7	1.571e-7	1.579e-5	7.320e-6	5.78e-2
	10Y	5.174e-7	5.337e-8	1.620e-6	6.130e-5	6.46e-4

The beta parameter exhibits no trend across swaption swap tenors nor across swaption expiries. A large proportion of the beta parameters are close to 0, indicating that for most of the swaptions, the DD model places a higher weightage on the arithmetic Brownian motion component. Despite this, the DD model was unable to closely match the market volatility, being unable to capture skew or smile effects of implied volatility. Hence, many of the market implied volatilities were either below or above the estimation derived from the DD model.

2.2 SABR Model

The SABR model was calibrated by fitting it to market swaption implied volatilities. The α, ρ, and ν parameters were determined by minimizing the error between the SABR model's lognormal implied volatilities and the observed market volatilities. Key parameters obtained are as follows:

Table 5: Alpha(α) Parameters

		Tenors				
		1Y	2Y	3Y	5Y	10Y
Start at	1Y	0.001391	0.001847	0.001969	0.001781	0.001710
	5Y	0.001666	0.001995	0.002103	0.001902	0.001748
	10Y	0.001782	0.001963	0.002083	0.002016	0.001802

Alpha determines the overall level of the IV estimated from the SABR model (the level of the volatility smile curve produced). The alpha parameter generally increases slightly with swaption expiry. This is consistent with an upward sloping implied volatility term structure.

Additionally, looking at the general profile of swaption ATM vols (shown in figure 2 below) one would expect the Alpha parameter of the top left and intermediates to have the highest alpha values. From the results however, the region with the highest Alpha parameters span the swaptions just below Top Left (5Y x 2Y) to the swaptions in the Intermediates (10Y x 5Y). This could possibly be due to the market preparing for a market-wide event in the intermediate term that traders are preparing for.

Figure 2: Swaptions expected to have higher ATM IVs

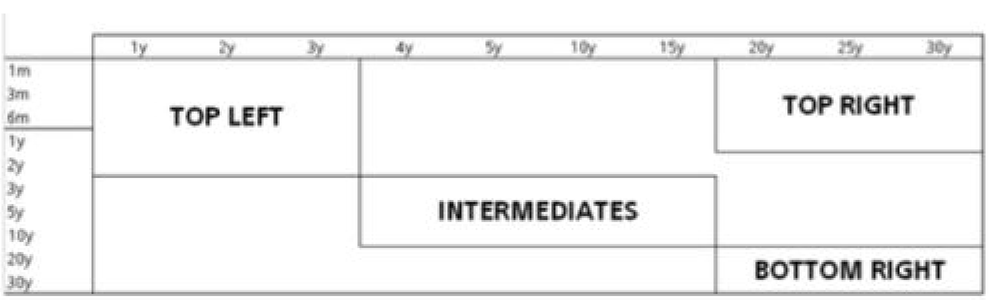


Table 6: Rho(ρ) parameters

Start at	Tenors: 1Y	2Y	3Y	5Y	10Y
1Y	-0.006323	-0.005251	-0.004828	-0.004145	-0.002557
5Y	-0.005857	-0.005470	-0.005497	-0.005068	-0.004157
10Y	-0.005484	-0.005475	-0.005536	-0.005637	-0.005084

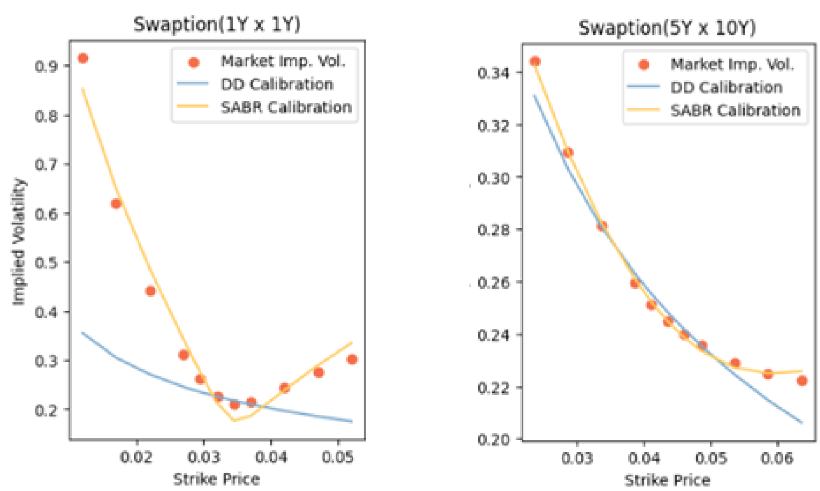
The Rho parameter measures the correlation between the forward rate and its volatility. It influences skewness of the volatility smile, with more negative rho values resulting in higher implied volatilities for OTM receiver swaptions relative to OTM payer swaptions. It generally increases (becomes less negative) with swaption swap tenor. However, this is less noticeable in the swaptions with 10Y to expiry. Rho also generally decreases (becomes more negative) with swaption expiry, indicating more pronounced skewness for options with longer time to expiry, with the notable exception of swaptions with 1Y tenor.

Table 7: Nu(v) parameters

Start at	Tenors: 1Y	2Y	3Y	5Y	10Y
1Y	0.02049	0.01677	0.01438	0.01065	0.00792
5Y	0.01340	0.01062	0.00937	0.00675	0.00511
10Y	0.01010	0.00929	0.00872	0.00723	0.00580

The Nu parameter measures the volatility of volatility. It controls kurtosis of the volatility smile, determining the curvature. It decreases as swaption swap tenor increases and generally decreases as swaption expiry increases. This indicates that the implied volatility smile decreases with increasing maturity, resulting in flatter smiles over time.

Figure 3: 1Y x 1Y Swaption with 0.02 Nu, 5Y x 10y swaption with 0.005 Nu



From figure 3 above, options that have a low Nu parameter such as the 5y x 10y and 10y x 10y, the DD model can match the implied volatilities to a reasonably accurate degree due to the market implied volatilities not having much smile effect, allowing the DD model to reasonably accurately fit the lower strikes. However, it will always be inaccurate for the higher strikes where the smile effect is more pronounced.

2.3 Swaptions Pricing under DDM vs SABR Model

The 2y x 10y payer swaptions and 8y x 10y swaptions were then priced by the Black 76 model with outputs obtained from the DD Model and the SABR model, with the relevant parameters from above.

Table 8: Swaption Pricings

	Payer swaptions (2y x 10y)		Receiver swaptions (8y x 10y)	
Strikes	**DD Model**	**SABR**	**DD Model**	**SABR**
1%	0.2881	0.2897	0.0190	0.0197
2%	0.1949	0.1984	0.0339	0.0390
3%	0.1123	0.1151	0.0566	0.0616
4%	0.0513	0.0520	0.0890	0.0905
5%	0.0174	0.0215	0.1321	0.1304
6%	0.0041	0.0110	0.1861	0.1865
7%	0.0007	0.0070	0.2506	0.2581
8%	0.0001	0.0050	0.3240	0.3396

Insights

The pricing comparison between the DDM and the SABR model offers valuable insights into the differences in how each model captures market-implied volatility smiles. As observed in Table 8, swaption prices generated by the SABR model are generally higher than those of the DD model, particularly in deep OTM regions, such as for high-strike payer swaptions and low-strike receiver swaptions. This difference reflects the SABR model's feature of stochastic volatility and skewness parameters, allowing it to more accurately fit the pronounced volatility smile typically observed in swaption markets.

Conversely, the differences are minimal near the money, where both models tend to converge. This suggests that while DDM can provide reasonable approximations for near-the-money pricing, the SABR model is more suitable for pricing structured products or options that depend on tail behaviour. These findings demonstrate the importance of model selection based on the use case

Part III – Convexity Correction

$$V_0 = D(0,T)g(F) + h'(F)[V^{\text{pay}}(F) - V^{\text{rec}}(F)] + \int_0^F h''(K)V^{\text{rec}}(K)\,dK + \int_F^\infty h''(K)V^{\text{pay}}(K)\,dK$$

Static replication of the CMS payoffs was carried out using the above equation was carried out with the following parameters:

$$g(K) = \mathrm{K}, \qquad g'(K) = 1, \qquad g''(K) = 0$$

$$h(K) = \frac{K}{\mathrm{IRR}(K)}$$

$$h'(K) = \frac{\mathrm{IRR}(K) - K * \mathrm{IRR}'(K)}{\mathrm{IRR}(K)^2}$$

$$h''(K) = \frac{-K * \mathrm{IRR}''(K) - 2 * \mathrm{IRR}'(K)g(K)}{\mathrm{IRR}(K)^2} - \frac{-2 * \mathrm{IRR}'(K)^2 * K}{\mathrm{IRR}(K)^3}$$

Specifics of the calculation are detailed below:

- The relevant discount factors were calculated using the respective OIS discount factors from the bootstrapped curve in Part 1.
- The SABR model calibrated in Part 2 was used to calculate the implied volatility used together with the Black 76 valuation formulas in the calculation of the receiver swaptions and the payer swaptions for the integral components.

3.1 Pricing Constant Maturity Swap

Table 9: CMS PVs

CMS	PV of Receiver Legs
CMS10y over the next 5 years (semi-annual)	0.20640
CMS2y over the next 10 years (quarterly)	0.38825

The 2y CMS is more expensive than the 10y CMS due to more frequent compounding and the longer duration of the swap.

3.2 Forward Swap Rates vs CMS Rates

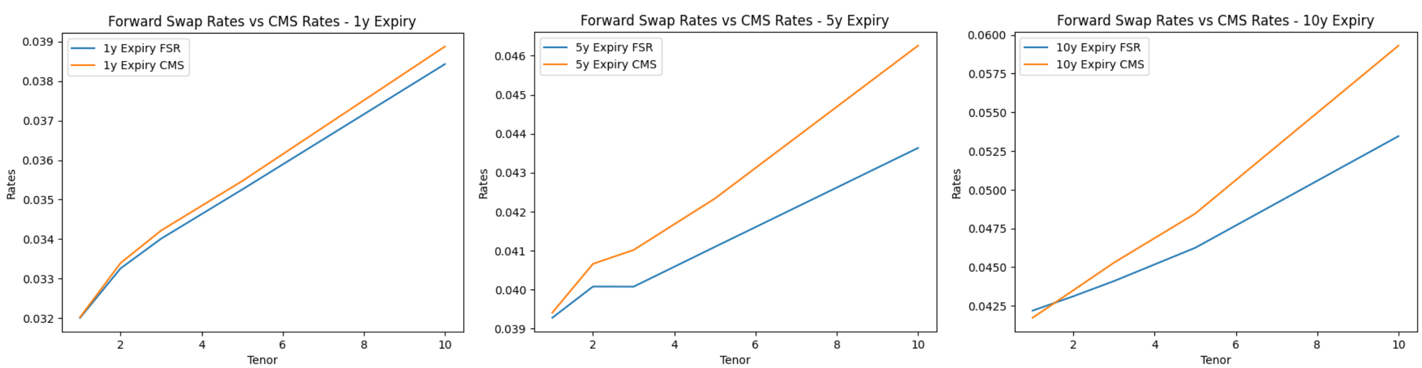
Table 10: Constant Maturity Swap Rates (%)

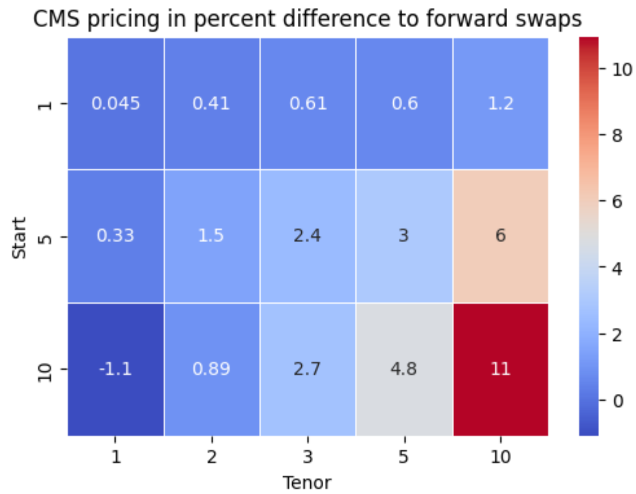
	Tenors	1Y	2Y	3Y	5Y	10Y
Start at	1Y	3.20	3.34	3.42	3.55	3.89
	5Y	3.94	4.07	4.10	4.23	4.63
	10Y	4.17	4.35	4.53	4.85	5.93

Table 11: Forward Swap Rates (%)

	Tenors	1Y	2Y	3Y	5Y	10Y
Start at	1Y	3.20	3.33	3.40	3.53	3.84
	5Y	3.93	4.01	4.01	4.11	4.36
	10Y	4.22	4.31	4.41	4.62	5.35

Figure 4: Comparisons between forward swap rates and CMS rates





Insights

Firstly, the CMS rates are consistently higher than their corresponding forward swap rates, a reflection of the positive convexity embedded in CMS instruments. This spread arises due to the non-linear relationship between the value of a swap and interest rates—specifically, the curvature in the pricing function resulting from the volatility of future rates.

Secondly, the spread between the forward swap rates and the CMS rates increases as maturity increases. We observe that for shorter expiries, such as the 1y × n swaps, the convexity effect is relatively muted, with differences remaining below 2% across all tenors. However, as the expiry extends into the 5y × n and 10y × n structures, the spread becomes significantly larger, reaching up to 3%–11% for longer-tenor instruments. This widening differential confirms that the magnitude of the convexity correction grows not just with the tenor of the underlying swap, but also with the time to expiry.

This has practical implications for pricing and hedging CMS-linked products (i.e., CMS caps/floors) which are highly sensitive to convexity effects. Ignoring convexity can lead to mispricing, particularly for long-dated instruments, and can result in significant P&L mismatches. The use of models like SABR, which account for volatility smiles and rate uncertainty, becomes crucial in accurately adjusting for this convexity premium.

Part IV – Decompounded Options

4.1 Static Replication (Part 1)

Using the method by Breeden and Litzenberger, we carried out static replication by expressing the payoff as the following formula with receiver and payer swaptions.

$$V_0 = D_0(0,T)g(F) + \int_0^F h''(K)V^{rec}(K)dK + \int_F^\infty h''(K)V^{pay}(K)dK$$

F = *Forward Swap Rate*
$D_O(0,T)$ = *OIS Discount Factor from time 0 to time T*
$g(F)$ = *Non-linear payoff function* $0.04^{1/q}$
$V^{rec}(K)$ = *Price of Receiver CMS Swaption with Strike K*
$V^{pay}(K)$ = *Price of Payer CMS Swaption with Strike K*

$$g(F) = F^{\frac{1}{4}} - 0.04^{\frac{1}{2}}, \qquad g'(F) = \frac{1}{4}F^{-\frac{3}{4}}, \qquad g''(F) = -\frac{3}{16}F^{-\frac{7}{4}}$$
$$h(K) = \frac{g(K)}{\mathrm{IRR}(K)}$$
$$h'(K) = \frac{\mathrm{IRR}(K)g'(K) - g(K)\mathrm{IRR}'(K)}{\mathrm{IRR}(K)^2}$$
$$h''(K) = \frac{\mathrm{IRR}(K)g''(K) - \mathrm{IRR}''(K)g(K) - 2\mathrm{IRR}'(K)g'(K) + 2\mathrm{IRR}'(K)^2 g(K)}{\mathrm{IRR}(K)^3}$$

Where $\mathrm{Payoff} = \mathrm{CMS}_{10y}^{1/p} - 0.04^{1/q}$ at time $\mathrm{T} = 5\mathrm{y}, \mathrm{p} = 4,$ and $\mathrm{q} = 2$. Additionally, we utilized the 5y x 10y Forward Swap Rate and OIS discount factor $D_0(0{,}5y)$from Part 1.

The present value of the contract is **0.2497**.

4.2 Static Replication (Part 2) CMS Caplet

Given another $\mathrm{Payoff} = \left(\mathrm{CMS}\ 10y^{\frac{1}{p}} - 0.04^{\frac{1}{q}}\right)^+$ at time $T = 5y$, where $p = 4$ and $q = 2$, this has now been reformulated as an option on CMS with strike L.

This is equivalent to a CMS caplet with nonlinear payoff $g(K)$ evaluated previously in 4.1 where $K = L$.

$$\therefore (F^{1/4} - 0.04^{1/2})^+ = (F^{1/4} - 0.2)^+$$
$$F^{1/4} - 0.2 > 0 \Rightarrow F > 0.2^4 = 0.0016 = L$$

Applying Breeden-Litzenberger formula we can derive the following,

$$CMS\ Caplet = D_0(0,T)\int_L^{\infty} g(K)f(K)dK = \int_L^{\infty} h(K)\frac{\partial^2 V^{pay}(K)}{\partial K^2}\,dK$$
$$= h'(L)V^{pay}(L) + \int_L^{\infty} h''(K)V^{pay}(K)\,dK$$

Utilizing the SABR parameters previously obtained in Part II and the valuation formula, the present value of this contract is **0.0312**. Notably, the CMS caplet-style has a lower PV than the first CMS-linked payoffs is intuitive given that the payoff only activates if the CMS exceeds a certain threshold, adding optionality and reducing expected value.

Concluding Insights

This project has deepened our understanding of fixed income modelling by applying advanced valuation and calibration techniques in a deliberately structured and market-consistent way. We explored how interest rate derivatives are priced and managed in practice, where the use of multi-curve frameworks, stochastic volatility models, and static replication are necessary to achieve precision valuation.

Throughout, we made conscious modelling choices—selecting discount curves based on collateral assumptions, calibrating volatility models to reflect observed market smiles, and valuing exotic payoffs through transparent, replication-based methods. These decisions reflect the broader industry shift toward precision, consistency, and risk-aware modelling, especially in the post-LIBOR and collateralized trading environment.